

Measurement of Cross Sections for the Indirect Production of ^{149}Tb via $^{151}\text{Eu}(^6,^7\text{Li},x)$ Reactions

Laura A. Bills^{a,b,*}, Alan B. McIntosh^{b,**}, Jonathan T. Morrell^c, Philip Adsley^{b,d}, Austin D. Abbott^{a,b}, Arthur Alvarez^{b,d}, Ryan Amberger^{b,d}, Tiara Anderson^{b,e}, Molly C. DeLuca^f, Jeremias Garcia-Duarte^g, Matthew D. Gott^{h,1}, Brooklyn D. Green^{a,b}, Kris Hagel^b, Travis Hankins^{a,b}, Jason T. Harke^g, Richard O. Hughes^g, Trenton Irvinⁱ, Lauren A. McIntosh^b, Yonatan Mishnayot^{g,2}, Connor Mohs^h, Claus Müller-Gattermann^h, Sharanya Palit^{b,j}, Sebastian Regener^{b,k}, Leonith Rodriguez^{b,l}, Aaron Salinas^{b,m}, John Santucci^{b,d}, Steven J. Schultz^{a,b}, Maxwell Sorensen^{a,b}, Aaron S. Tamashiro^g, Justin Tobar^{a,b}, C. Etienne Vermeulenⁿ, Benjamin Wellons^{b,d} and Sherry J. Yennello^{b,d,***}

^aDepartment of Chemistry, Texas A&M University, College Station, Texas, 77843, USA

^bCyclotron Institute, Texas A&M University, College Station, Texas, 77843, USA

^cBrookhaven National Laboratory, Upton, New York, 11973, USA

^dDepartment of Physics & Astronomy, Texas A&M University, College Station, Texas, 77843, USA

^eDepartment of Physics, Florida State University, Tallahassee, Florida, 32306, USA

^fDepartment of Medical Physics, University of Wisconsin-Madison, Madison, Wisconsin, 53705, USA

^gLawrence Livermore National Laboratory, Livermore, California, 94550, USA

^hPhysics Division, Argonne National Laboratory, Lemont, Illinois, 60439, USA

ⁱDepartments of Radiology and Chemistry, University of Alabama at Birmingham, Birmingham, Alabama, 35233, USA

^jDepartment of Physics, University of Dallas, Irving, Texas, 75062, USA

^kDepartment of Mechanical Engineering, Texas A&M University, College Station, Texas, 77843, USA

^lDepartment of Physics, The University of Texas at El Paso, El Paso, Texas, 79968, USA

^mDepartment of Physics, Texas Lutheran University, Seguin, Texas, 78155, USA

ⁿLos Alamos National Laboratory, Los Alamos, New Mexico, 87545, USA

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ABSTRACT

Terbium-149 is an ideal candidate for targeted alpha therapy (TAT) as it decays both via alpha emission and positron emission, making it a theranostic isotope. However, production of this isotope is often a challenge, in part due to the production of its excited isomeric state, $^{149\text{m}}\text{Tb}$, which primarily does not decay into the ground state. Therefore, production of this isomer decreases the overall production of the TAT candidate, $^{149\text{g}}\text{Tb}$. In an effort to avoid production of this isomer, indirect production reactions focusing on the direct production of other isotopes such as ^{149}Er , ^{149}Ho , and ^{149}Dy are attractive alternatives, as all of these isotopes decay with short half-lives directly into $^{149\text{g}}\text{Tb}$. For the first time, we are reporting cross sections for reactions using ^6Li and ^7Li beams on ^{151}Eu targets with goal of producing ^{149}Dy to accumulate larger quantities of $^{149\text{g}}\text{Tb}$. Comparing these results to previously measured direct production mechanisms, we can conclude that higher energy indirect production reactions can be used to achieve equivalent or greater production rates of $^{149\text{g}}\text{Tb}$, when compared with lower energy direct reactions. These methods could also allow for more flexibility of the target material, making them more accessible. Finally, this work demonstrates the ratio of production of the ground state to metastable state isomer production remains constant for increasing energies among the energy range that was measured, and is overall much higher than those measured in the direct production reactions.

1. Introduction

As targeted alpha therapy (TAT) becomes more popular as a cancer treatment modality [1–4], it is useful to find and develop production capabilities for the most optimal radionuclides required for this treatment. Some of these

candidate nuclides, such as ^{211}At and ^{225}Ac , have well-known production mechanisms [4–8], while others still face large production challenges. Terbium-149 ($t_{1/2} = 4.12$ h) is one of the nuclides that has been identified as a possible candidate for TAT due to its 16.7% α -decay branch [9, 10]. Additionally, it is especially unique compared with the other identified TAT nuclides in that it also decays via positron emission, allowing it to also be imaged via positron emission tomography (PET). This capability makes ^{149}Tb one of the only possible theranostic TAT nuclides, without the need for a separate imaging agent. Since Tb also has a stable isotope, ^{159}Tb , some preliminary research can also be explored without the use of radioactive materials, increasing safety and

*Corresponding author

**Co-corresponding author

***Principal Investigator

✉ lamccan@tamu.edu (L.A. Bills)

ORCID(s):

¹Present Address: Enrichment Science and Engineering Division, Oak Ridge National Laboratory, Oak Ridge, TN 37830, USA

²Present Address: Soreq Nuclear Research Center, Yavne, 818000, Israel

accessibility, in addition to other radioactive isotopes being explored for imaging, such as ^{152}Tb and ^{155}Tb .

While currently ^{149}Tb is most commonly produced via spallation on a tantalum target [11–13], this method is not widely available and alternative production mechanisms are still under investigation. Previous work has focused on numerous production mechanisms for this nuclide via light-ion [14–18] and heavy-ion [19–22] direct production reactions, but the competing production of the metastable isomeric state of ^{149}Tb , $^{149\text{m}}\text{Tb}$, which primarily decays to ^{149}Gd rather than $^{149\text{g}}\text{Tb}$, has hindered overall production cross sections. In order to maximize the yield of the TAT candidate, $^{149\text{g}}\text{Tb}$, indirect production methods have instead been proposed [20]. These productions involve focusing on the direct production of nuclides higher in the decay chain such as ^{149}Er ($t_{1/2} = 4$ s), ^{149}Ho ($t_{1/2} = 21$ s), and ^{149}Dy ($t_{1/2} = 4.2$ min) which rapidly decay into only $^{149\text{g}}\text{Tb}$, increasing the total yield of this isotope. Building on previous experiments that reported ratios greater than 10 for the $^{149\text{m}}\text{Tb}$ to $^{149\text{g}}\text{Tb}$ ratio for ^6Li -induced direct production reactions [23, 24], this experiment proposes utilizing the same beams to indirectly produce $^{149\text{g}}\text{Tb}$ through the decay of ^{149}Dy by irradiating targets of ^{151}Eu rather than Sm. The energy ranges chosen for these experiments were based on the PACE4 reaction model [25, 26], and were selected to be 85, 90, and 95 MeV for the ^6Li beams and 105 and 110 MeV for the ^7Li beams. These experiments were chosen to determine the effect of these types of productions on the overall ratio of production of $^{149\text{g}}\text{Tb}$ to $^{149\text{m}}\text{Tb}$, and discover a larger range of potential production mechanisms that might be useful for maximizing the total production yield of $^{149\text{g}}\text{Tb}$ for TAT.

2. Methods

2.1. Target Preparation

The metallic ^{151}Eu targets used for the ^7Li reactions were prepared according to the physical vapor deposition method described in Bills et al. [23]. The remaining targets irradiated in the ^6Li measurements were made using the molecular deposition procedure described for the $^{152}\text{Gd}_2\text{O}_3$ targets in Bills et al. [24]. The enriched ^{151}Eu was acquired from ISOFLEX USA with 99.16% enrichment. All of the targets were produced with areal densities between 600–1000 $\mu\text{g}/\text{cm}^2$ so that the beam energy was not degraded significantly through the target.

2.2. Experimental Setup and Data Analysis

The experimental setup was identical to the one described in Bills et al. [24, 27], using the Hyperion HPGe detector array [28] to measure the targets immediately following irradiation, with offline single-crystal HPGe detectors used for longer counting hours to days following the irradiation. The Curie python package [29] was used for a majority of the data analysis, including the simultaneous peak fitting of all of the measured nuclides. Figure 1 represents a part of a γ -spectrum from 0–1000 keV measured by the summed

4 clovers (16 total detectors) that were utilized from the Hyperion detector array for the first 10 min following the irradiation of ^{151}Eu by a ^6Li beam at 95 MeV. The peaks for $^{149\text{m}}\text{Tb}$ and $^{149\text{g}}\text{Tb}$ are clearly identified, while ^{149}Dy was unable to be measured directly due to its small branching ratios for its γ -transitions (<16%), the most intense of which overlap directly with other measured nuclides.

3. Results and Discussion

Although these reactions were performed with the aim of measuring the overall yield of $^{149\text{g}}\text{Tb}$, cross sections for a total of 20 different nuclides and nuclidic states were able to be measured. In order to provide a complete picture of the reaction products and provide a complete dataset for all of the measured nuclides, cross sections were calculated for all possible nuclides, with results reported in Table 1. Since some of the Dy isotopes have short half-lives and complicated decay schemes, not all of the cross sections that were able to be calculated are independent cross sections, and those marked with an * are cumulative cross sections for that nuclide representing a summed total cross section for that nuclide and all parent nuclides in the decay chain. For example, the independent cross section for ^{149}Dy was unable to be calculated, as such the cross sections for $^{149\text{g}}\text{Tb}$ are all reported as cumulative cross sections as they are sums of both the $^{149\text{g}}\text{Tb}$ and ^{149}Dy independent cross sections. These cumulative cross sections also provide an upper limit on the parent cross sections, such as ^{149}Dy in this example. Since this experiment was designed to calculate the total amount of $^{149\text{g}}\text{Tb}$ produced both independently and as a result of this decay, these cumulative cross sections are the values necessary for comparison of these indirect production reactions to previous direct production reactions.

The resulting cross sections presented in Table 1 were also compared with a series of common predictive reaction models (PACE4 [25, 26], ALICE 2017 [30], and EMPIRE [31]) to assess their predictive capabilities. These comparisons are shown in Figure 2 for all of the measured Tb isotopes. Since PACE4 cannot distinguish between ground and excited isomeric state and cross sections for all of the Dy isotopes were unable to be determined, these plots compare a summed cumulative cross section of all three of these states when possible. As ^{154}Dy ($t_{1/2} = 2.75 \times 10^6$ y) is sufficiently long-lived, the measured and predicted production of ^{154}Tb was assumed to be independent.

Comparing the three reaction codes with each other, the differences in the predicted peaks of the cross sections for each isotope vary widely from one model to the next. For example, ALICE predicts the peak of the ^{149}Tb cross section around 90 MeV, EMPIRE predicts this peak around 115 MeV, and PACE4 predicts around 100 MeV. Since the models vary significantly from one another in energies, comparing the trends indicated by the measured Tb cross sections to this data is extremely valuable in determining where the peak of the excitation function actually lies. PACE4 appears to have the most accurate energy prediction for a majority of the

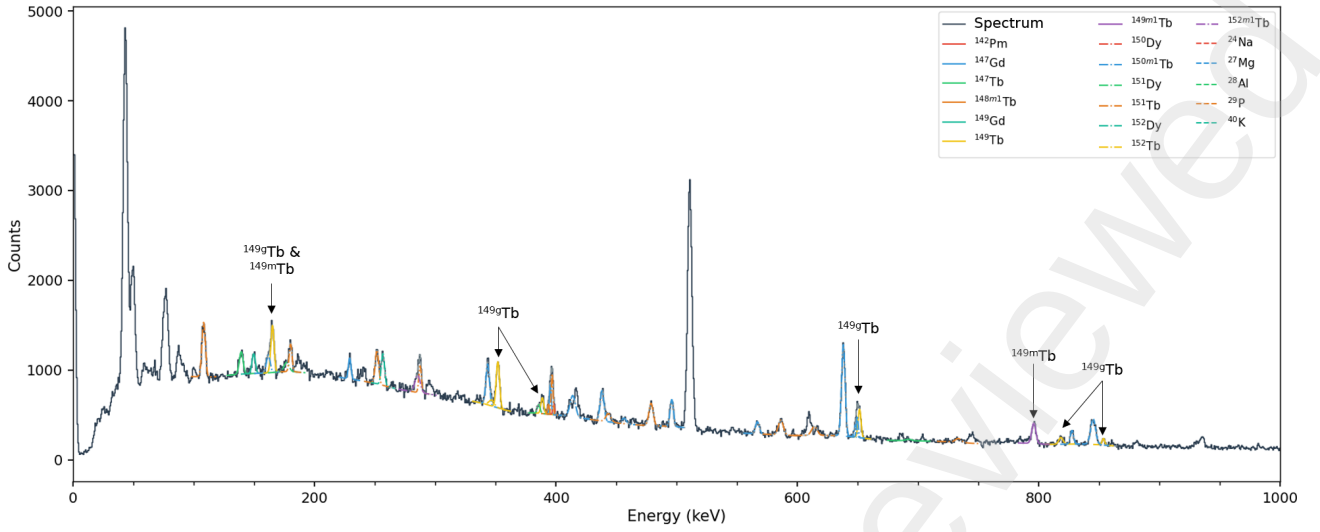


Figure 1: Gamma spectrum of the first 10 min post-irradiation of a ^{151}Eu by a ^6Li beam at 95 MeV measured by the summed Hyperion detectors. The colored peak fits show many of the measured nuclides able to be seen during this time window, with $^{149\text{m}}\text{Tb}$ and $^{149\text{g}}\text{Tb}$ peaks clearly identified.

	$^{151}\text{Eu}(^6\text{Li}, x)$ @ 85 MeV	$^{151}\text{Eu}(^6\text{Li}, x)$ @ 90 MeV	$^{151}\text{Eu}(^6\text{Li}, x)$ @ 95 MeV	$^{151}\text{Eu}(^7\text{Li}, x)$ @ 105 MeV	$^{151}\text{Eu}(^7\text{Li}, x)$ @ 110 MeV
^{146}Eu	8.5 ± 1.5	14 ± 5	26 ± 4	34 ± 8	51 ± 9
^{147}Eu	31 ± 5	30 ± 9	21 ± 3	57 ± 15	68 ± 16
^{148}Eu	22 ± 2	—	15 ± 2	—	—
^{146}Gd	$109 \pm 12^*$	$75 \pm 13^*$	$89 \pm 13^*$	$133 \pm 48^*$	—
^{147}Gd	$107 \pm 12^*$	—	$61 \pm 10^*$	$161 \pm 20^*$	$158 \pm 24^*$
$^{147\text{g}}\text{Tb}$	$3.3 \pm 1.0^*$	$75 \pm 24^*$	$11 \pm 4^*$	$14 \pm 4^*$	$18 \pm 5^*$
$^{148\text{g}}\text{Tb}$	—	$26 \pm 10^*$	$10 \pm 3^*$	—	—
$^{149\text{g}}\text{Tb}$	$31 \pm 3^*$	$51 \pm 6^*$	$55 \pm 6^*$	$65 \pm 7^*$	$82 \pm 9^*$
$^{149\text{m}}\text{Tb}$	88 ± 17	106 ± 38	138 ± 33	190 ± 32	192 ± 38
$^{150\text{g}}\text{Tb}$	$163 \pm 23^*$	$128 \pm 32^*$	$83 \pm 21^*$	$167 \pm 30^*$	75 ± 53
$^{150\text{m}}\text{Tb}$	259 ± 44	296 ± 83	245 ± 59	363 ± 95	411 ± 91
^{151}Tb	—	—	$165 \pm 17^*$	—	—
$^{152\text{g}}\text{Tb}$	$144 \pm 16^*$	$86 \pm 10^*$	$61 \pm 7^*$	62 ± 24	17 ± 8
$^{152\text{m}}\text{Tb}$	—	—	—	80 ± 41	112 ± 63
^{153}Tb	32 ± 14	19 ± 8	—	41 ± 11	—
^{154}Tb	5.1 ± 0.6	6.2 ± 0.9	4.0 ± 0.8	9.2 ± 1.4	12 ± 2
^{150}Dy	—	—	—	—	232 ± 117
^{151}Dy	375 ± 47	246 ± 75	—	397 ± 81	345 ± 143
^{152}Dy	43 ± 5	31 ± 4	15 ± 2	68 ± 8	55 ± 7
^{153}Dy	13 ± 11	8.2 ± 6.5	19 ± 5	34 ± 9	58 ± 18

Table 1

Measured cross sections (mb) for produced isotopes measured from the $^{151}\text{Eu}(^6\text{Li}, x)$ @ 85 MeV, $^{151}\text{Eu}(^6\text{Li}, x)$ @ 90 MeV, $^{151}\text{Eu}(^6\text{Li}, x)$ @ 95 MeV, $^{151}\text{Eu}(^7\text{Li}, x)$ @ 105 MeV, and $^{151}\text{Eu}(^7\text{Li}, x)$ @ 110 MeV reaction systems. Cross sections marked by * indicate cumulative production cross sections, while unmarked cross sections are independent cross sections.

isotopes, although there seems to be a far more significant overlap between each excitation function with less of an energy shift in between differing isotopes. In comparing the magnitudes of the cross sections, ALICE seems to be in the middle of the three models with its prediction accuracy, while seeming to overpredict a majority of the measured values. For the lower mass Tb isotopes ($^{147-152}\text{Tb}$), PACE4 seems to predict the magnitudes of these cross sections most accurately, with predictions for $^{148-150}\text{Tb}$ agreeing quite

well with the measured results. For the higher mass isotopes ($^{152-154}\text{Tb}$), EMPIRE seems to do the best job of predicting the magnitude, however, based on the major energy shift with respect to the measured data and overall significantly lower magnitude predicted cross sections of the lower mass isotopes, it is possible that the agreement is simply better since the cross sections measured for higher Z isotopes are lower due to being on the tails of the excitation functions. More measurements are necessary to fully confirm the peaks

of these excitation functions experimentally since only three energy points were measured in this work.

Since these measurements are the first indirect production methods for ^{149}Tb where both ^{149g}Tb and ^{149m}Tb were both able to be measured, it is useful to compare the ratio of production of these states to determine whether or not the indirect production mechanism is actually effective at maximizing the ^{149g}Tb to ^{149m}Tb production ratio, thereby potentially producing higher yields of the TAT radionuclide, ^{149g}Tb . In Figure 3, the ratio of production of ^{149g}Tb to ^{149m}Tb for previous direct production systems using Li beams on ^{149}Sm targets is compared to the indirect production of the same isomers with the same projectiles on the ^{151}Eu targets measured here. The ratios measured by the direct production reactions are represented as circles, while the indirect production reactions are represented by diamonds. For the systems where ^6Li is the projectile, the points are closed, and hollow for ^7Li . In this figure, it can be seen that the direct production measurements show a slight decrease in the production rate of ^{149g}Tb to ^{149m}Tb at higher energies, which is expected given the higher overall energy of the compound nucleus. In the indirect production reactions, it can be seen that the ratio of production of these 2 states is relatively constant with respect to system energy, indicating that the production of the parent isotope of ^{149g}Tb rather than the direct production likely increases the overall yield of ^{149g}Tb with respect to ^{149m}Tb . This is consistent with expectations since ^{149m}Tb can only be produced independently, while the production of ^{149g}Tb is a summation of its independent production and the decay of the production of ^{149}Dy . Additionally, Figure 3 also shows significantly larger ratios for the indirect production of ^{149g}Tb with respect to ^{149m}Tb than via the direct production, potentially indicating higher yields of ^{149g}Tb .

In order to better compare which light-ion production reaction is most effective for producing ^{149g}Tb , Figure 4 shows the saturation yields for all of the previously measured light-ion production reactions, as calculated for a 1 mg/cm² target thickness. Since there is a limited energy range of data available for some of the measured reactions, a very thin target with minimal energy loss (< 0.25 MeV) through the target was chosen for this comparison. As shown in Figure 4, the indirect production reactions represented by the red diamonds, demonstrate yields consistent with those produced by the direct production via ^3He on the same target [15]. This is promising in that it shows the potential for producing equivalent or higher yields using the indirect production of ^{149g}Tb than previously able to achieve via a more limited number of direct production reactions. While the proton direct production on ^{152}Gd still represents the highest yield [18, 24], it is important to measure other systems with greater target accessibility, since ^{152}Gd only has a 0.2% natural abundance, limiting the available target enrichment to only 30.6% and making it very costly for routine irradiations. It is also noted that these proposed reactions utilize a Li-projectile, which is less widely accessible compared to proton and He beams, in addition to having

lower available beam currents for larger production quantities. This work incites additional mechanisms using more widely available beam species to be explored, focusing on the indirect production of ^{149g}Tb through direct production of its parent isotopes.

4. Conclusions

This work measures for the first time over 75 new cross sections for nuclides/isomeric states produced via ^6Li - and ^7Li -induced reactions on ^{151}Eu targets, with the intention of demonstrating the potential value of indirect production modalities for the significant production of ^{149g}Tb for TAT. Through this work, the measurements of the reaction products produced in these reactions have been reported to help inform predictive models and future measurements. Additionally, these indirect production mechanisms have been confirmed to produce higher ratios of the production of ^{149g}Tb to non-medically relevant ^{149m}Tb , as compared to previously measured direct production reactions. Finally, the overall saturation yields for ^{149g}Tb produced by these indirect reactions were shown to be equivalent to other light-ion induced accelerator productions of ^{149g}Tb . These results show promise for other indirect production reactions of this nuclide potentially producing larger quantities than currently available via other methods.

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CRediT authorship contribution statement

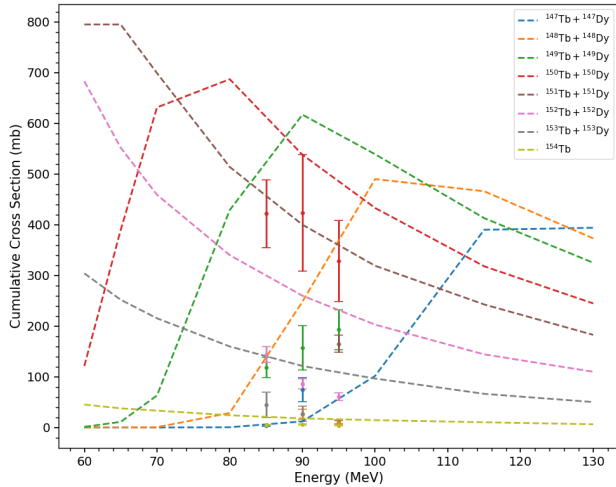
Laura A. Bills: Conceptualized the study; prepared and conducted the experiment; analyzed experimental results; drafted and revised the manuscript. **Alan B. McIntosh:** Conceptualized the study; prepared and conducted the experiment; analyzed experimental results; revised the manuscript. **Jonathan T. Morrell:** Analyzed experimental results; revised the manuscript. **Philip Adsley:** Prepared

and conducted the experiment; analyzed experimental results; revised the manuscript. **Austin D. Abbott:** Conducted the experiment; revised the manuscript. **Arthur Alvarez:** Conducted the experiment; revised the manuscript. **Ryan Amberger:** Prepared and conducted the experiment; revised the manuscript. **Tiara Anderson:** Conducted the experiment; revised the manuscript. **Molly C. DeLuca:** Conducted the experiment; revised the manuscript. **Jeremias Garcia-Duarte:** Conceptualized the study; revised the manuscript. **Matthew D. Gott:** Prepared the experiment; revised the manuscript. **Brooklyn D. Green:** Conducted the experiment; revised the manuscript. **Kris Hagel:** Prepared and conducted the experiment; analyzed experimental results; revised the manuscript. **Travis Hankins:** Conducted the experiment; revised the manuscript. **Jason T. Harke:** Conceptualized the study; revised the manuscript. **Richard O. Hughes:** Conceptualized the study; revised the manuscript. **Trenton Irvin:** Conducted the experiment; revised the manuscript. **Lauren A. McIntosh:** Conceptualized the study; revised the manuscript. **Yonatan Mishnayot:** Conceptualized the study; revised the manuscript. **Connor Mohs:** Prepared the experiment; revised the manuscript. **Claus Müller-Gattermann:** Prepared the experiment; revised the manuscript. **Sharanya Palit:** Conducted the experiment; revised the manuscript. **Sebastian Regener:** Conducted the experiment; revised the manuscript. **Leonith Rodriguez:** Conducted the experiment; revised the manuscript. **Aaron Salinas:** Conducted the experiment; revised the manuscript. **John Santucci:** Prepared and conducted the experiment; revised the manuscript. **Steven J. Schultz:** Conducted the experiment; revised the manuscript. **Maxwell Sorensen:** Conducted the experiment; revised the manuscript. **Aaron S. Tamashiro:** Conceptualized the study; analyzed experimental results; revised the manuscript. **Justin Tobar:** Conducted the experiment; revised the manuscript. **C. Etienne Vermeulen:** Conceptualized the study; prepared and conducted the experiment; revised the manuscript. **Benjamin Wellons:** Conducted the experiment; revised the manuscript. **Sherry J. Yennello:** Conceptualized the study; prepared the experiment; analyzed experimental results; revised the manuscript.

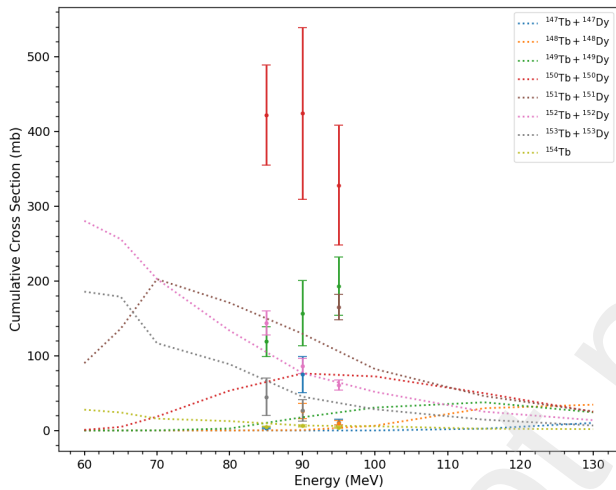
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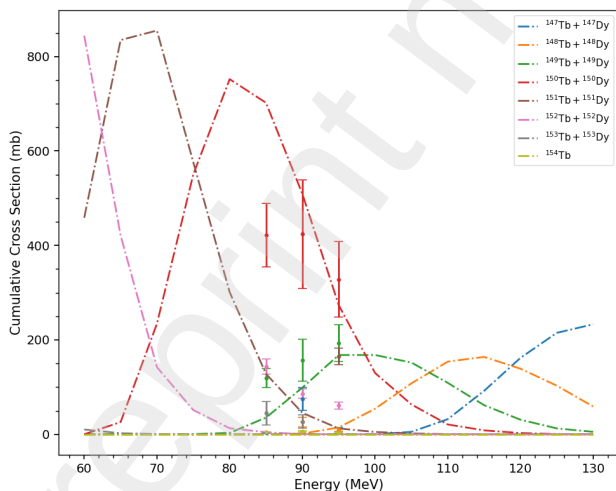
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(a)



(b)



(c)

Figure 2: Measured cumulative cross sections for the Tb isotopes (sum of ground state, metastable state, and Dy parent) from the $^{151}\text{Eu}(^6\text{Li},x)$ reaction system compared with (a) ALICE (dashed), (b) EMPIRE (dotted), and (c) PACE4 (dot-dashed) predictions.

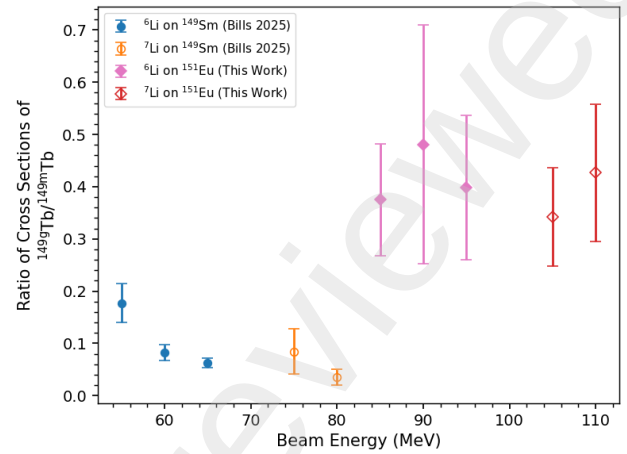


Figure 3: Ratio of the cross sections for ^{149g}Tb to ^{149m}Tb for the direct and indirect production of ^{149}Tb using varying Li beams. The circular points represent the direct production reactions [23, 24, 27], and the diamond points the indirect production routes (this work). ^6Li beams are represented by the solid points while the ^7Li beams are hollow. The different colors help differentiate between the 4 different reactions being compared.

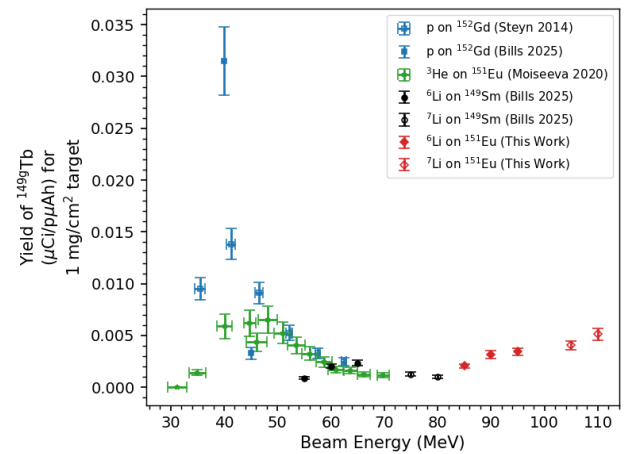


Figure 4: Saturation production yields in $\mu\text{Ci}/\text{p}\mu\text{Ah}$ for ^{149g}Tb calculated for $1 \text{ mg}/\text{cm}^2$ targets. Proton-induced direct productions are shown by the blue squares [18, 24], He-induced direct productions by the green points [15], Li-induced direct productions by the black circles [23, 24], and the indirect Li-induced reactions by the red diamonds (this work).